

Notes: Rosenzweig & Udry (RES, 2020)

External Validity in a Stochastic World: Evidence from Low-Income Countries

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1 Big picture

- **Question.** How externally valid are internally valid estimates of treatment effects or investment returns when a population faces time-varying aggregate shocks?
- **Core argument.** A single-period causal estimate can be internally valid for the realized state of the world, but not representative of the expected return over the distribution of future states.
- **Key distinction.** The paper separates *sampling uncertainty* from *state-of-the-world uncertainty*. Standard errors usually capture only the former.
- **Empirical settings.** The paper studies investment returns in India agriculture, Ghana agriculture, Sri Lankan micro-enterprises, and Indonesian schooling.
- **Main shocks.** Rainfall, rainfall timing, output prices, oil prices, exchange rates, and seasonality alter returns.
- **Main result.** Returns fluctuate substantially over time, and confidence intervals based only on sampling error are much too narrow.
- **External-validity message.** A study conducted in one year tells us about that year, but policy makers typically need expected returns over many possible future states.
- **Methodological takeaway.** Evaluation designs should collect information on relevant aggregate shocks and estimate how treatment effects interact with those shocks.
- **Policy takeaway.** Policies based on one-period RCTs or single-year return estimates may be misleading in low-income economies where weather and commodity shocks are large.

2 Data and measurement

2.1 India ICRISAT villages, 2005–2011

- The India analysis uses agricultural panel data from four ICRISAT villages in Maharashtra.
- The key investment is planting-stage input spending: ploughing, seeding, fertilizer, labour, and material inputs.
- The main aggregate state is Kharif-season rainfall.
- The identification strategy uses IMD pre-planting monsoon forecasts interacted with farmer and plot characteristics as instruments for planting-stage investment.

Interpretation: Farmers react to forecasts before rainfall is realized. This creates variation in investment that can identify returns conditional on the eventual rainfall state.

2.2 Northern Ghana villages, 2009–2012

- The Ghana analysis uses panel data on 1,352 households in 75 communities in northern Ghana.
- The key investment is planting-stage investment, linked to realized seasonal rainfall and a rainfall timing index.
- Randomized offers of rainfall insurance and cash grants provide instruments for planting investment.
- The data contain geographically localized rainfall variation across eleven clusters and multiple years.

Interpretation: The Ghana data allow the paper to test temporal variation and whether spatial rainfall variation can substitute for time-series variation.

2.3 Sri Lankan micro-enterprises, 2004–2006

- The paper reuses the De Mel, McKenzie, and Woodruff micro-enterprise experiment.
- Randomized cash and capital grants identify returns to non-land investment.
- The main aggregate variation is seasonal demand variation linked to the local agricultural cycle.
- The dry season, from December to March, is expected to depress demand for non-farm enterprise output.

Interpretation: Even non-farm enterprise returns can be affected by agricultural shocks because local demand depends on farm income.

2.4 Urban Indonesia schooling returns, 1993–2000

- The Indonesia analysis uses IFLS panel data on urban male wage workers.
- The outcome is log hourly wages; the key investment is years of schooling.
- Macro shocks are oil prices and the rupiah exchange-rate shock during the Asian financial crisis.
- The paper asks whether schooling returns vary year by year and whether oil and exchange-rate shocks explain that variation.

Interpretation: Schooling returns matter for long-horizon education decisions, but a one-year wage return estimate may reflect temporary commodity or exchange-rate shocks.

3 Models and empirical specifications

3.1 Investment choice under aggregate shocks

An agent chooses an investment a_{i0} before a future aggregate state s_1 is realized:

$$\max_{a_{i0} \in A} E_{s_1} [V_{i1} (\pi(a_{i0}, s_1, \lambda_i, \varepsilon_i), z_i) \mid \Omega_{i0}]. \quad (1)$$

Interpretation: The investment decision is based on the entire distribution of possible future states, not only on the state realized in the researcher's sample year.

3.2 Expected marginal return for a risk-neutral agent

$$0 = \int \varepsilon_i \frac{\partial y_1(a_{i0}, s_1)}{\partial a_{i0}} f(s_1 \mid \Omega_{i0}) ds_1 - 1. \quad (2)$$

Interpretation: The expected marginal product of investment, averaged over aggregate states, determines the optimal investment level.

3.3 State-specific return observed by the researcher

$$\beta_i(a_{i0}, \varepsilon_i, s_1) = \varepsilon_i \frac{\partial y_1(a_{i0}, s_1)}{\partial a_{i0}} - 1. \quad (3)$$

Interpretation: A single-year estimate is a return conditional on one aggregate state. It is not automatically the expected return over the state distribution.

3.4 Linear aggregate-shock interaction

The state-specific return can be summarized as

$$\beta(s_1) = \beta_0 + \alpha'(s_1 - E[s_1]). \quad (4)$$

Interpretation: The vector α measures how returns respond to aggregate shocks. Estimating α is the key to improving temporal external validity.

3.5 Why conventional confidence intervals are too narrow

If the aggregate shock is omitted, a researcher reports sampling uncertainty but not future state uncertainty:

$$CI_{policy} = CI_{sampling} + \text{uncertainty from future aggregate states}. \quad (5)$$

Interpretation: Conventional standard errors can be internally correct for the observed sample but externally misleading for future implementation.

3.6 Omitted shock result

With two aggregate shocks, if one dimension is unobserved, measured variance is too small:

$$\sigma_T^2 - \sigma_M^2 = \alpha_b^2(1 - \rho^2). \quad (6)$$

Interpretation: Any omitted aggregate shock that is not perfectly correlated with the included shock causes external-validity uncertainty to be understated.

3.7 Empirical specifications

- **India:** farmer fixed effects IV profit functions with planting-stage investment, investment squared, rainfall, and soil interactions.
- **Ghana:** farmer fixed effects IV profit functions with planting investment interacted with seasonal rainfall and rainfall timing.
- **Sri Lanka:** enterprise fixed effects IV regressions of net returns on randomized investment grants and seasonal interactions.
- **Indonesia:** OLS and worker fixed effects log wage regressions with year-specific schooling returns and macro-shock interactions.

Interpretation: All specifications ask whether returns vary with measurable aggregate states.

4 Identification and empirical design

4.1 What identifies the main claim

- The paper combines internally valid sources of variation in investment with observed aggregate shocks.

- The goal is not merely to estimate one return, but to estimate how the return moves with aggregate conditions.
- Panel data over multiple years are essential because aggregate states change over time.

Interpretation: The identification target is richer than a conventional treatment effect: it is a distribution of effects across states.

4.2 Why cross-sectional shock variation is risky

- Rainfall realizations in one year can be correlated with site characteristics such as long-run rainfall means and household wealth.
- Cross-sectional variation cannot reliably separate shock realization from unobserved spatial heterogeneity.
- The Ghana evidence shows that cross-sectional estimates do not consistently replicate time-series relationships.

Interpretation: Spatial variation can help, but only when the researcher can credibly control for spatial unobservables or use repeated time variation.

4.3 Limits of the design

- Only selected aggregate shocks are observed; unobserved shocks still widen true policy-relevant uncertainty.
- Agricultural returns may also depend on pests, disease, credit conditions, input prices, and policy interventions.
- The Indonesian wage analysis identifies macro-linked variation in returns, but education decisions are long-term and depend on expected future returns.

Interpretation: The estimates should be read as lower bounds on the external-validity problem.

5 Tables and figures with full interpretation

5.1 Table 1: Farmer fixed effects IV estimates of planting-stage investments on net returns, four ICRISAT villages, 2005–11; and Figure 1: Estimated returns by rainfall realization, four ICRISAT villages

Read:

- Table 1 estimates how planting-stage investment affects net returns in the ICRISAT villages.
- The mean investment effect is positive, around 6.4 to 6.6 at sample mean investment and rainfall.
- Rainfall interactions imply that returns are not constant across states of the world.
- Figure 1 shows the predicted return increasing with Kharif rainfall.

Interpretation: The table establishes the first key empirical ingredient: investment returns are sensitive to rainfall. The figure translates regression coefficients into a state-contingent return schedule. A single estimate obtained in a low-rainfall year would imply a much lower return than one obtained in a high-rainfall year.

Economic message: Farmers face a technology in which the payoff to planting investment depends heavily on realized weather. Internal validity in one rainfall state does not solve external validity across rainfall states.

What not to over-read: The upward slope does not mean every farmer should always invest more. Risk, credit constraints, soil heterogeneity, and expectations all matter for the ex ante decision.

Variable	1	2
Planting-stage investment	7.15 (2.47)	1.54 (4.09)
Planting-stage investment squared $\times 10^{-5}$	-7.26 (3.19)	-3.91 (4.60)
Planting-stage investment $\times$ Kharif rainfall	—	0.016; (0.006)
Planting-stage investment squared $\times$ Kharif rainfall $\times 10^{-5}$	—	-0.012; (0.006)
Planting-stage investment $\times$ Kharif rainfall $\times$ depth 1	—	-0.010; (0.003)
Planting-stage investment $\times$ Kharif rainfall $\times$ depth 3	—	-0.002; (0.004)
Planting-stage investment $\times$ Kharif rainfall $\times$ sandy soil	—	0.002; (0.003)
Planting-stage investment $\times$ Kharif rainfall $\times$ loam soil	—	0.004; (0.007)
H_0 : Investment effect = 0, $\chi^2(n)$ [p]	11.1(2) [0.0039]	14.3(4) [0.006]
Investment effect at sample mean investment and rainfall	6.41 (2.16)	6.63 (2.90)
N (farmer-years)	1,152	1,152

Notes: Standard errors clustered at the farm level in parentheses. Specification includes rainfall and lagged rainfall; rainfall and lagged rainfall interacted with total landholdings, irrigated landholdings, and four soil characteristics; ; village*year fixed effects. The identifying instruments for the planting-stage investment variables are the year-spec IMD Southern Peninsula pre-planting monsoon forecast interacted with total landholdings, irrigated landholdings, ;

Table 1

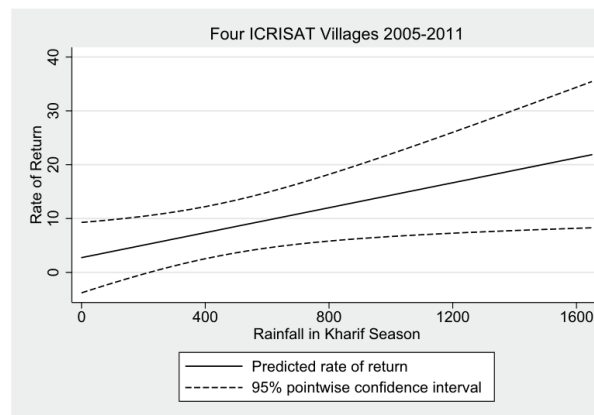


FIGURE 1
Estimated returns by rainfall realization: four ICRISAT villages 2005–11

Figure 1

5.2 Figure 2: Distribution of returns and weather shocks, India; and Table 2: Probability of being within $\pm\delta$ of expected return

Read:

- Figure 2 compares distributions based on rainfall variation, sampling variation, and both together.
- Rainfall-inclusive distributions are much wider than sampling-only distributions.
- Table 2 reports very low probabilities that a single-year return is close to the expected return.

- For India, the probability of being within 10 percentage points of the expected return is about 1.2%; even within 50 percentage points it is only about 6%.

Interpretation: Conventional standard errors answer how precisely the return was estimated in the realized year. The paper asks how close that realized-year return is to the expected return across future states. That second question has much wider uncertainty.

Economic message: Policy makers need the distribution of returns across weather states, not just cross-sectional precision in one state.

What not to over-read: Table 2 probabilities are lower-bound calculations because they include rainfall but not every possible aggregate shock.

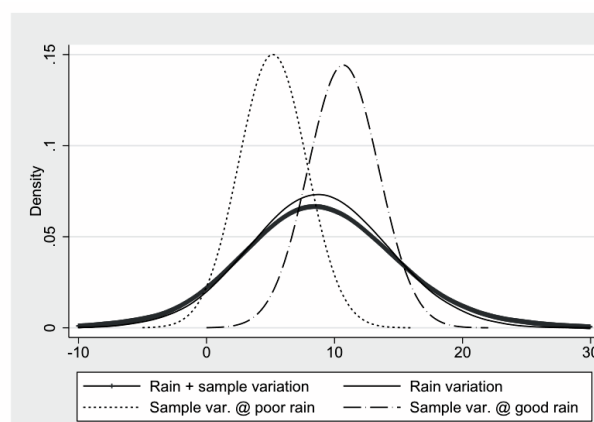


Figure 2

δ (percentage points)	ICRISAT	Ghi
10	0.012	0
20	0.02	0
30	0.04	0
40	0.05	0
50	0.06	0

Notes: The probabilities are calculated based on equations (2.3) and (2.4) in the text. In column 1, we use the estimate α (the interaction between investment and the rainfall realization, evaluated at mean investment) from Table 1, column multiplied by the estimated standard deviation of the rainfall distribution in the ICRISAT village with the longest rainfall series. In column 2, we use the estimates of α from column 2 of Table 3, evaluated at the mean investment level, an empirical distribution of the bivariate weather outcomes in the historical rainfall distribution of the Ghana study area.

Table 2

5.3 Figure 3: Returns to planting-stage investment, eleven geographic clusters of Ghana villages; and Table 3: Farmer fixed effects IV estimates in Ghana

Read:

- Figure 3 shows large variation in cluster-year returns, with raw estimates, shrinkage estimates, and confidence intervals.
- Table 3 estimates Ghana returns using randomized cash and insurance offers as instruments.
- Rainfall timing and total rainfall interactions are jointly significant in the longer panel specification.

Interpretation: Ghana provides a second agricultural setting with different instruments and data. The same pattern emerges: returns vary substantially across time and space. Shrinkage estimates show that some apparent variation is sampling noise, but real variation remains large.

Economic message: Macro-state dependence is not an India-specific artifact. It appears in another agricultural environment with randomized instruments.

What not to over-read: Figure 3's raw cluster-year estimates should not be treated as exact local treatment effects; they are noisy and require shrinkage interpretation.

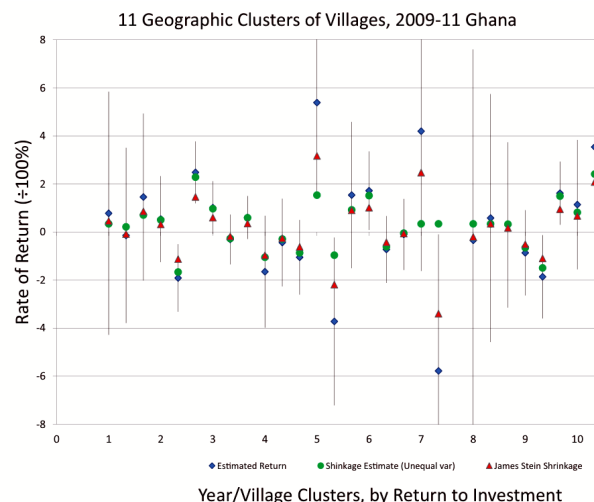


Figure 3

TABLE 3
Farmer fixed effects IV estimates of planting-stage investments on net returns, 75 Ghana Villages, 2009-11

Sample years	2009-11		2009-10
Variable	1	2	3
Planting-stage investment	1.043 (0.707)	0.956 (0.718)	-0.011 (0.764)
Planting-stage investment $\times$ rainfall index	-	0.0298 (0.0119)	0.031 (0.017)
Planting-stage investment $\times$ total rainfall	-	0.123 (0.0586)	0.121 (0.058)
H_0 : Investment effect = 0, χ^2 (3) [p]	-	11.3 [0.0104]	7.84 [0.048]
H_0 : Rainfall-investment interactions = 0, χ^2 (2) [p]	-	6.32 [0.0424]	7.78 [0.021]
N (farmer-plot-years)	6,418	6,418	4,126

Notes: Standard errors clustered at the farmer level in parentheses. The specifications include the rainfall index its square, five soil characteristics, five slope characteristics, these characteristics interacted with the rainfall index, distance of the plot to the house, the number of trees, total rainfall, total rainfall and the rainfall index interacted, area, and year. The identifying instruments for the planting-stage variables are randomized offers of index insurance, nine price levels, cash grants, and cash grants plus insurance.

Table 3

5.4 Figure 4: Estimated returns by rainfall realization in Ghana; and Figure 5: Distribution of returns and weather shocks, Ghana

Read:

- Figure 4 shows predicted returns rising with both seasonal rainfall and rainfall timing index.
- Figure 5 shows that rainfall variation generates a wide distribution of returns.
- Sampling-only distributions at good and poor rainfall states are much narrower and shifted relative to one another.

Interpretation: Figure 4 shows the mechanism: rainfall changes the return to investment. Figure 5 shows the inference implication: the return estimate depends on the rainfall state realized. Together they show why single-year studies can be internally valid and externally fragile.

Economic message: A recommendation based on a good-rain year may be inappropriate in a bad-rain year.

What not to over-read: The figures isolate rainfall-related variation; they do not imply rainfall is the only relevant aggregate state.

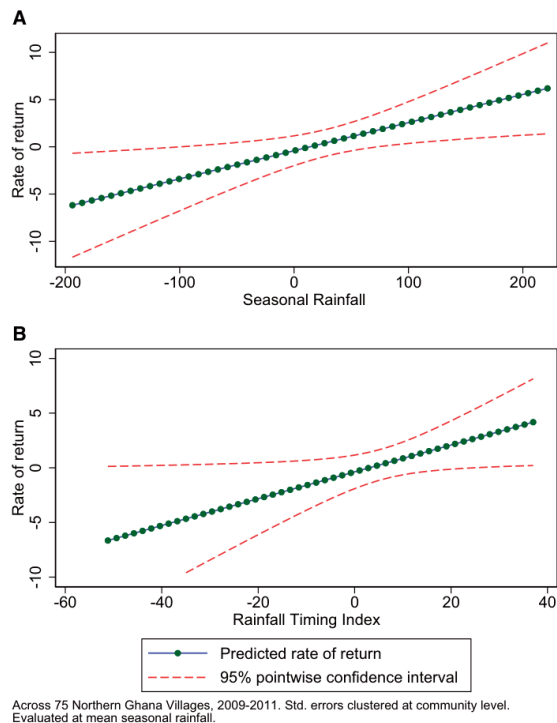


FIGURE 4
Figure 4

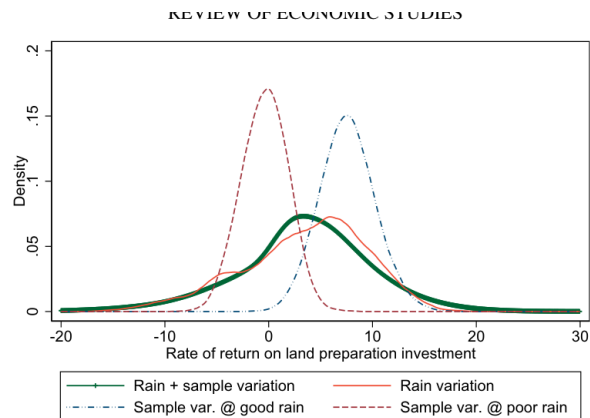


Figure 5

5.5 Table 4: Cross-sectional IV estimates in Ghana; and Figure 6: Panel versus cross-sectional rainfall relationships

Read:

- Table 4 estimates returns separately using cross-sectional variation in each year.
- Rainfall interactions vary substantially by year and sometimes disagree with panel estimates.
- Figure 6 compares cross-sectional relationships with the panel relationship.

Interpretation: This section tests a tempting shortcut: use spatial variation in a shock to proxy for temporal variation. The results show that this can be unreliable because places with different rainfall realizations can also differ in wealth, soil, infrastructure, and long-run rainfall distributions.

Economic message: Spatial variation is not a free substitute for time-series variation. External validity requires knowing whether the cross-section really mimics the future state distribution.

What not to over-read: The authors are not saying cross-sectional designs are always invalid. They show that the identifying assumption is strong and questionable here.

TABLE 4
Cross-sectional IV estimates of planting-stage investments on net returns, by year 75 Ghana Villages, 2009–11

Variable/sample	2009	2010	2011
Planting-stage investment	1.402 (6.27)	0.720 (0.950)	1.715 (1.80)
Planting-stage investment $\times$ rainfall index	0.0391 (0.0645)	-0.00121 (0.0118)	0.0702 (0.0296)
Planting-stage investment $\times$ total rainfall	0.176 (0.0858)	0.0544 (0.0826)	0.169 (0.127)
H0: Investment effect = 0, χ^2 (3) [p]	6.28 [0.0990]	1.54 [0.6737]	11.4 [0.0097]
H0: Rainfall-investment interactions = 0, χ^2 (2) [p]	4.62 [0.0993]	0.450 [0.7979]	5.64 [0.0597]
N (farmer-plot-years)	1,844	2,282	2,292

Notes: Standard errors clustered at the farmer level in parentheses. The specifications include the rainfall index and its square, five soil characteristics, five slope characteristics, these characteristics interacted with the rainfall index, the distance of the plot to the house, the number of trees, total rainfall, total rainfall and the rainfall index interacted, plot area, and year. The identifying instruments for the planting-stage variables are randomized offers of index insurance: nine price levels, cash grants, and cash grants plus insurance.

Table 4

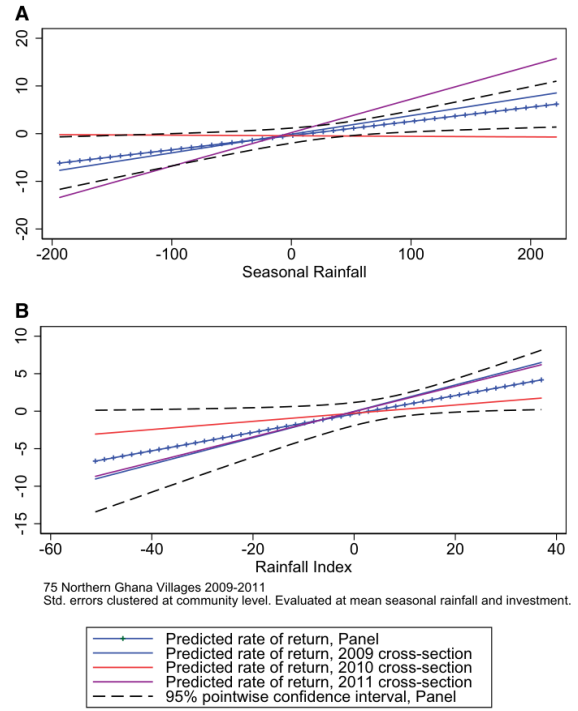


FIGURE 6

Figure 6

5.6 Table 5: Sri Lankan micro-enterprise returns by survey wave; and Table 6: Sri Lankan returns by quarterly season

Read:

- Table 5 shows that returns vary by survey wave, rejecting equality across waves.
- Table 6 groups returns by season and shows statistically meaningful seasonal variation.
- Returns are lowest in the January–March dry-quarter period, consistent with lower agricultural demand spilling into non-farm businesses.

Interpretation: The Sri Lanka case extends the argument beyond agriculture. Micro-enterprise returns also depend on aggregate local demand conditions. Even randomized grants to firms can have returns that vary with the season in which outcomes are measured.

Economic message: External validity problems apply to non-farm firms in agrarian economies because local demand is linked to farm income cycles.

What not to over-read: Seasonality explains some variation, not all variation. Other local shocks, firm heterogeneity, and measurement issues remain relevant.

TABLE 5
Enterprise-fixed effects IV estimates of investments on net returns, by survey wave Sri Lankan Micro-enterprises, 200

Variable	1	2
Investment (non-land)	0.0756 (0.0242)	—
Entrepreneur sick	−1,042.8 (329.4)	−1,033 (391.9)
Wave 2 investment	—	0.13 (0.07)
Wave 3 investment	—	0.13 (0.05)
Wave 4 investment	—	0.06 (0.03)
Wave 5 investment	—	0.07 (0.03)
Wave 6 investment	—	0.06 (0.02)
Wave 7 investment	—	0.07 (0.02)
Wave 8 investment	—	0.07 (0.02)
Wave 9 investment	—	0.08 (0.02)
Wave fixed effects	Y	Y
H ₀ : wave fixed effects = 0, χ^2 (8) [p]	30.1 [0.0002]	5.45 [0.70]
H ₀ : Seasonal pattern, χ^2 (4) [p]	—	2.22 [0.69]
N (enterprise-quarters)	3,101	3,101

Table 5

TABLE 6
Enterprise-fixed effects IV estimates of investments on net returns, by quarterly season Sri Lankan Micro-enterprise 2004–6

Variable	1	2
Investment (non-land)	—	0.097 (0.021)
Entrepreneur sick	−1,066.5 (366.5)	−951.5 (345.3)
Q1 investment	0.0947 (0.0219)	—
Q2 investment	0.1001 (0.0236)	—
Q3 investment	0.0693 (0.0192)	—
Q4 investment	0.0836 (0.0242)	—
Season fixed effects	Y	Y
H ₀ : Season fixed effects = 0 χ^2 (4) [p]	9.27 [0.0547]	11.6 [0.020]
N (enterprise-quarters)	3,101	3,101

Notes: Standard errors clustered at the enterprise level in parentheses. Sample excludes enterprises directly affected by 2004 tsunami. The identifying instruments in the first column are the randomized offer of cash and that variable interact with whether the entrepreneur was a male and those two instruments interacted with wave indicators. The identifying instruments in column 2 are the randomized offer of cash and that variable interacted with whether the entrepreneur was a male and quarter indicators.

Table 6

5.7 Figure 7: Oil price and exchange rate fluctuations in Indonesia; Table 7: Year-specific schooling returns; and Table 8: Schooling returns and external macro shocks

Read:

- Figure 7 shows major oil-price and exchange-rate shocks in Indonesia during the 1990s.
- Table 7 shows that returns to schooling vary significantly across years.
- Table 8 links variation to oil prices and exchange-rate shocks interacted with industry openness and skill intensity.

Interpretation: The Indonesia analysis shows that the external-validity problem is not limited to physical investment. Human-capital returns also vary with macro shocks. If a schooling-return estimate is measured in a boom or crisis year, it may not represent the expected return faced by students making long-term education decisions.

Economic message: Commodity and exchange-rate shocks change relative demand for skilled labour, altering returns to schooling.

What not to over-read: The results do not imply that education is unprofitable. They imply that one-year return estimates are not stable structural constants.

FIGURE 7 OIL PRICE AND EXCHANGE RATE FLUCTUATIONS IN INDONESIA, 1990–2000

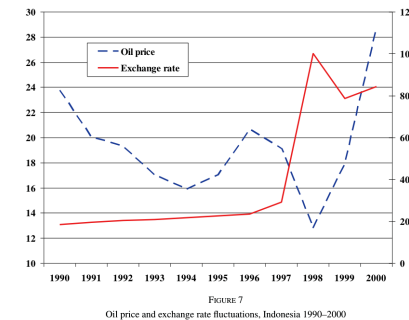


Figure 7

TABLE 7
Log urban hourly wages: estimates of the returns to schooling by year, 1993–2000 men aged 16–79 in urban Indonesia (IFLS 1993, 1997, 2000)

Variable/estimation procedure	OLS	OLS	FE
Years of schooling	0.123 (0.00485)	—	—
Years of schooling/difference from 2000, 1993	0.154 (0.0103)	−0.003 (0.010)	—
Years of schooling/difference from 2000, 1994	0.124 (0.00730)	−0.012 (0.101)	—
Years of schooling/difference from 2000, 1995	0.121 (0.00692)	−0.013 (0.010)	—
Years of schooling/difference from 2000, 1996	0.122 (0.00677)	−0.012 (0.010)	—
Years of schooling/difference from 2000, 1997	0.138 (0.00768)	−0.000 (0.010)	—
Years of schooling/difference from 2000, 1998	0.104 (0.00653)	−0.025 (0.007)	—
Years of schooling/difference from 2000, 1999	0.102 (0.00625)	−0.027 (0.007)	—
Years of schooling 2000	0.129 (0.00761)	—	0.007
Year fixed effects	Y	Y	Y
H ₀ : Year fixed effects=0, F(8, 3062) [p]	415.0 [0.0000]	395.1 [0.0000]	17.0 [0.0000]
H ₀ : Schooling returns are equal across years, F(7, 3,062) [p]	5.94 [0.0000]	2.95 [0.0000]	0.004

Table 7

TABLE 8
Log urban hourly wages: estimates of the returns to schooling and external macro shocks men aged 16–79 in urban Indonesia (IFLS 1993, 1997, 2000)

Sample period	1993–2000	1998–2000
Estimation procedure	OLS	FE
Years of schooling	0.0855 (0.0102)	−0.00134 (0.0213)
Years of schooling × oil price	0.00105 (0.000492)	0.00156 (0.000434)
Years of schooling × exchange rate × industry openness × ind. skill × 10 ^{−3}	—	0.222 (0.0506)
Oil price	−0.0528 (0.0136)	−0.0504 (0.0106)
Exchange rate × industry openness × skill × 10 ^{−4}	—	−0.0884 (0.0131)
Exchange rate × 10 ^{−4}	—	−0.447 (0.107)
Industry openness × skill	—	0.0968 (0.0535)
H ₀ : No shock effects on schooling returns, F(2, 2,120) [p]	12.242 [0.0001]	12.242 [0.0001]
N (person-years)	5,470	5,470

Notes: Standard errors clustered at the individual level in parentheses. Specifications also include age and age squared year, and age and age squared interacted with the oil price/industry variables.

Table 8

5.8 Figure 8: Average annual OPEC crude oil price, 1960–2016

Read:

- Figure 8 shows large oil-price variation over several decades.
- Years relevant to major Indonesian schooling studies are marked.
- The oil-price environment in any one study year may be unusually low or high relative to the longer-run distribution.

Interpretation: This figure links the paper’s empirical estimates to the broader problem of evaluating influential single-period schooling studies. A return estimate obtained in one oil-price state may not generalize to other oil-price states.

Economic message: Long-run macro time series can be used to ask whether a study year was typical or atypical.

What not to over-read: Oil prices are not the only determinant of schooling returns; they are one measurable macro state that illustrates the method.

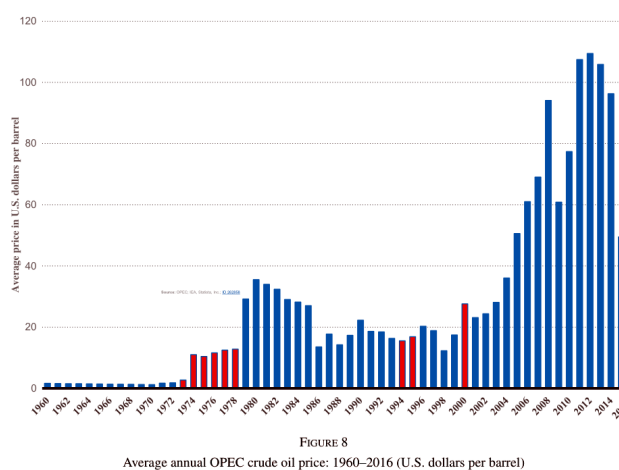


Figure 8

5.9 Appendix Figure A1: Output price variation; Figure A2: Predicted and realized Ghana returns; and Table A1: Rainfall and baseline wealth

Read:

- Figure A1 shows that output prices fluctuate substantially in Ghana and India.
- Figure A2 compares predicted and realized Ghana returns; rainfall shocks are useful but incomplete predictors.
- Table A1 shows rainfall and mean rainfall are related to baseline wealth, supporting concerns about cross-sectional rainfall variation.

Interpretation: The appendix deepens the identification point. Price variation is another measurable source of state dependence. Predictions based on rainfall help but do not perfectly forecast returns. Baseline wealth correlations show why cross-sectional estimates can be contaminated by long-run spatial differences.

Economic message: External validity requires both shock time series and an understanding of how those shocks are correlated with local characteristics.

What not to over-read: Appendix predictions demonstrate improvement over ignoring shocks, not full external validity.

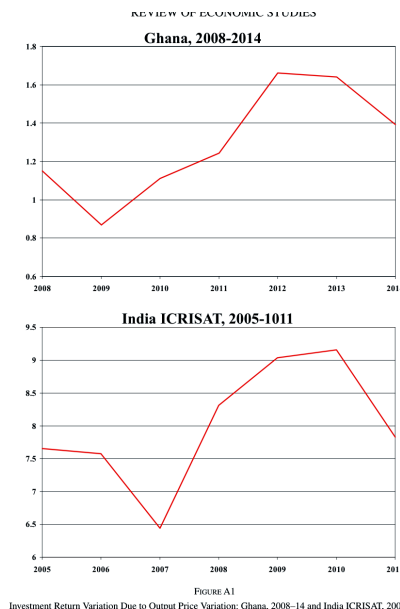


Figure A1

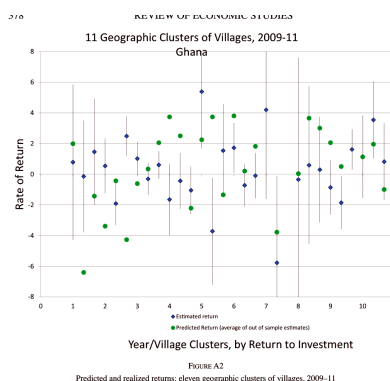


Figure A2

TABLE A1
Estimates of the effects of year-specific rainfall and mean rainfall on log total household per-capita wealth at harvest (2009)

Variable/rainfall years	2000–2015	2009	2010	2011
Rainfall in year	–	–0.731 (0.986)	11.8 (4.35)	0.0193 (0.00565)
Rainfall in year squared	–	0.0568 (0.0295)	–0.767 (0.285)	–0.000011 (0.000003)
Mean rainfall	0.263 (0.0193)	–	–	–
Mean rainfall squared	–0.0000122 (0.0000122)	–	–	–
Number of farmers	1,065	1,065	1,065	1,065
H: No rainfall effect $F(2, 68)$ [p]	4.70 (0.0123)	4.62 (0.0132)	3.89 (0.0251)	10.37 (0.0001)

Note: Standard errors clustered at the village level in parentheses.

Table A1

6 Short summary

- The paper studies temporal external validity in environments subject to aggregate shocks.
- Single-year estimates can be internally valid but externally fragile.
- Conventional confidence intervals usually include sampling error but not state-of-the-world variation.
- India and Ghana agricultural returns vary strongly with rainfall and rainfall timing.
- The probability that a one-year agricultural return estimate lies close to the expected return is extremely low in the paper's calculations.
- Sri Lankan micro-enterprise returns vary by season, reflecting agricultural demand spillovers.
- Indonesian schooling returns vary by year and interact with oil-price and exchange-rate shocks.
- Cross-sectional rainfall variation does not reliably replicate time-series rainfall variation.
- Appendix evidence shows that output prices and wealth-rainfall correlations further complicate external validity.

7 Conclusion

7.1 Core conclusion

Rosenzweig and Udry show that external validity in a stochastic world is fundamentally a distributional problem. The relevant policy object is not only the causal effect in one realized state, but the distribution of effects across future aggregate states.

7.2 Why the paper matters for RCTs and causal inference

The paper does not reject randomized or quasi-experimental evidence. It argues that internal validity is only the first step. When aggregate shocks change returns, a one-time experiment estimates a state-contingent effect.

7.3 Economic intuition

Investment decisions are made before uncertain states are realized. Agents and policy makers care about expected returns and risk over the entire state distribution. If rainfall, prices, or exchange rates change returns, the payoff to investment is a function of the state of nature.

7.4 Implications for policy

Policies based on single-period estimates can be overconfident. A fertilizer subsidy, cash grant, micro-enterprise programme, or schooling intervention may have high returns in one state and low returns in another. The right evaluation should report not only the estimate and sampling standard error, but also how the effect changes with relevant aggregate shocks.

7.5 Implications for future research design

- Collect data over multiple periods whenever aggregate shocks are important.
- Measure economically relevant aggregate states such as rainfall, prices, demand, seasonality, and exchange rates.
- Estimate interactions between treatment effects and aggregate shocks.
- Use historical shock distributions to compute policy-relevant confidence intervals.
- Be cautious about using cross-sectional spatial variation as a substitute for time-series variation.

7.6 Final takeaway

External validity is not merely about whether one population resembles another. It is also about whether the state of the world in the study year resembles the distribution of states in which the policy will operate. In stochastic environments, credible policy evaluation requires both identification and a model of the shocks.